

Digital Fundamentals - Conversion

Date

No

Decimal $\rightarrow$ Hexa

$$109_{10} \rightarrow \begin{array}{l} 109 \div 16 = 6 \text{ R } 13 \\ 6 \div 16 = 0 \text{ R } 6 \end{array} \uparrow 6D_{16}$$

$$219_{10} \rightarrow \begin{array}{l} 219 \div 16 = 13 \text{ R } 11 \\ 13 \div 16 = 0 \text{ R } 13 \end{array} \uparrow DB_{16}$$

$$853_{10} \rightarrow \begin{array}{l} 853 \div 16 = 53 \text{ R } 5 \\ 53 \div 16 = 3 \text{ R } 5 \\ 3 \div 16 = 0 \text{ R } 3 \end{array} \uparrow 355_{16}$$

Hexa	Decimal
A	10
B	11
C	12
D	13
E	14
F	15

Hexa $\rightarrow$ Decimal

$$32A_{16} \rightarrow 10 + 32 + 768 = 810_{10}$$

$$\begin{array}{c} \downarrow \\ \downarrow \times 16^0 \\ \downarrow \times 16^1 \end{array}$$

$$\times 16^2$$

$$B3F_{16} \rightarrow 15 + 48 + 2816 = 2879_{10}$$

$$\begin{array}{c} \downarrow \\ \downarrow \times 16^0 \\ \downarrow \times 16^1 \end{array}$$

$$\times 16^2$$

$$C3D9_{16} \rightarrow 9 + 208 + 768 + 49152 = 50137_{10}$$

$$\begin{array}{c} \downarrow \downarrow \downarrow \\ \downarrow \times 16^2 \times 16^1 \times 16^0 \end{array}$$

$$\times 16^3$$

Hexa $\rightarrow$ Dec

$$\begin{array}{l} \nearrow 0.75 \quad \nearrow 0.015625 \\ 12 \times 16^{-1} \quad 4 \times 16^{-2} \\ \uparrow \quad \nearrow \end{array}$$

$$28B.C4_{16} = 11 + 128 + 512 + 0.75 + 0.015625 \\ = 691.765625_{10}$$

$$\begin{array}{l} \downarrow \\ 11 \times 16^0 = 11 \\ \downarrow \\ 8 \times 16^1 = 128 \\ \downarrow \end{array}$$

$$2 \times 16^2 = 512$$

Binary $\rightarrow$ Decimal

$$10110 \rightarrow 16 + 4 + 2 = 22$$

$$16 \ 8 \ 4 \ 2 \ 1$$

$$1011 \rightarrow 1 + 2 + 8 = 11$$

$$8 \ 4 \ 2 \ 1$$

$$11011 \rightarrow 1 + 2 + 8 + 16 = 27$$

$$16 \ 8 \ 4 \ 2 \ 1$$

Decimal $\rightarrow$ Binary

$$37 \rightarrow \begin{array}{r} 1 \ 0 \ 0 \ 1 \ 0 \ 1 \\ \hline 32 \ 16 \ 8 \ 4 \ 2 \ 1 \end{array}$$

$$26 \rightarrow \begin{array}{r} 1 \ 1 \ 0 \ 1 \ 0 \\ \hline 16 \ 8 \ 4 \ 2 \ 1 \end{array}$$

$$14 \rightarrow \begin{array}{r} 1 \ 1 \ 1 \ 0 \\ \hline 8 \ 4 \ 2 \ 1 \end{array}$$

$$35 \rightarrow \begin{array}{r} 1 \ 0 \ 0 \ 0 \ 1 \ 1 \\ \hline 32 \ 16 \ 8 \ 4 \ 2 \ 1 \end{array}$$

